
Machine Learning Based Analog Circuit Fault Diagnosis
PDF 1 — Chapters 1-4: Project Foundations
Overview | Problem Statement & Motivation | Objectives | Complete Methodology

CHAPTER 1 — Project Overview

Chapter 1 - Project Overview Before Starting This is not a normal book. This is exactly like our discussions. Whenever possible I'll ask questions, answer them, give examples, explain the code, explain the electronics, explain the ML and prepare you for viva simultaneously. Whenever you see

PROFESSOR ASKS

it means a question that can actually be asked in your viva. Whenever you see

EXCELLENT ANSWER

memorize the concept, not the wording.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer: [OK] What is this project? [OK] Why was it built? [OK] What problem are we solving? [OK] Why is the problem important? [OK] What is the complete workflow? [OK] Why machine learning? [OK] Why analog circuits? [OK] What is the final output? [OK] What are the major modules? [OK] Explain the project in 2 minutes. [OK] Explain the project in 10 minutes. [OK] Answer basic viva questions.

Difficulty

Basics 5/5

Theory 5/5

Coding 1/5

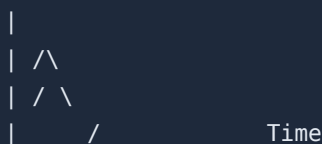
Viva Importance 5/5

Let's Begin

Suppose someone asks you "What is your project?" Don't immediately say Machine Learning Based Analog Circuit Fault Diagnosis. That's just the title. Titles don't explain projects. Instead answer like this.

Excellent Answer

Our project is an intelligent fault diagnosis system for analog electronic circuits. It automatically identifies the fault present inside a circuit by analyzing its transient voltage waveform using machine learning. Instead of manually looking at waveforms and guessing the fault, the computer predicts the fault automatically. Simple. Clear. Professional. Question Why do we even need this project? Good question. Let's understand. Imagine You are an engineer. A circuit stops working. You connect an oscilloscope. You get something like this Voltage



|
| / \
| / \
|_____/_____ Time

Question. Can you immediately tell whether resistor is open resistor is short capacitor is damaged inductor is faulty No. You need experience. Sometimes lots of experience. Now imagine 1000 circuits. Can one engineer manually inspect all waveforms? Almost impossible. **Need** — Automation. That is where our project begins.

Problem Statement

The problem is very simple. Finding faults inside analog circuits manually is slow difficult requires expert knowledge becomes harder as circuits become more complex Therefore we want a computer

to identify faults automatically.

One Line Definition

If someone asks "What exactly does your project do?" Answer It predicts the fault present inside an analog circuit from its transient response waveform using machine learning. That's it. One sentence.

Real World Importance

Many students skip this. Don't. Our project is useful in Communication Systems Medical Electronics Industrial Automation Power Electronics Embedded Systems Consumer Electronics Aerospace Electronics Automotive Electronics Basically everywhere analog circuits exist. Example Imagine a hospital ECG machine. Suppose one resistor becomes faulty. Machine starts giving wrong readings. Can we manually inspect every resistor? No. Instead automatic diagnosis can immediately identify the faulty component. Same idea. Different application.

Another Example

Imagine an aircraft. Hundreds of analog circuits. **Manual diagnosis** — takes hours. **Automatic diagnosis** — takes seconds.

PROFESSOR ASKS
Why is fault diagnosis important?
EXCELLENT ANSWER

Fault diagnosis improves reliability, reduces maintenance time, minimizes downtime, and helps detect failures before they become critical. Excellent.

PROFESSOR ASKS
Why analog circuits?
EXCELLENT ANSWER

Analog circuits are more difficult to diagnose than digital circuits because their outputs change continuously rather than producing simple logic levels. Very important.

PROFESSOR ASKS
Why not digital circuits?
EXCELLENT ANSWER

Digital circuits generally produce binary outputs, making fault detection comparatively simpler. Analog circuits exhibit continuous voltage and current variations, making diagnosis significantly more challenging. So what exactly are we building? Think of it like this. We are building an intelligent doctor. Not for humans. For circuits. **Patient** — Circuit. Symptoms

Voltage Waveform

Doctor

Machine Learning

Diagnosis

Fault Class

Very easy to remember.

Complete Project Flow

Everything in our project can be summarized using this diagram.

Analog Circuit

Healthy / Faulty

LTspice Simulation

Voltage Waveforms

Interpolation

Feature Extraction

17 Features

StandardScaler

Machine Learning

Prediction — Evaluation. Dashboard

Error Analysis

Improvement

Final Model

Congratulations. You just understood the complete project in one diagram. Don't Worry At this point you probably understand only 20%. That's okay. Every chapter will explain one box. By the end the entire diagram will become obvious.

Major Modules

Our project has seven major modules.

Module 1

Circuit Design

Design circuits inside LTspice.

Module 2

Dataset Generation

Generate healthy and faulty waveforms.

Module 3

Signal Processing

Interpolation Preprocessing

Feature Extraction

Module 4

Machine Learning

Train SVM (and later ANN)

Module 5

Evaluation Accuracy Precision Recall F1

Confusion Matrix

Module 6

Analysis

Feature Importance

Error Analysis

DC Level

Topology Rule

Module 7

Presentation Dashboard Report Viva These seven modules are the entire project. Why Machine Learning? Question. Can we simply write rules? Example

If Peak Voltage > 5V

RC Open Question. Will this work? Maybe. For one fault. What about 16 faults? And 17 features? Impossible. Need Machine Learning.

Machine Learning

learns the relationships automatically.

One Important Thing

Machine Learning

does NOT know electronics. It only knows numbers. For example Peak 5.2 RMS 3.8

Rise Time

2.1. DC Level 4.8

Machine Learning

only sees numbers. It is our job to convert the waveform into meaningful numbers. That process is called Feature Extraction. We'll study that later. Input Question. What goes inside the model? Not the circuit. Not the waveform. The input is

17 Features

Very important. Output Question. What comes out? One label. Example RC_Open or RL_Short or RLC_Normal One prediction.

PROFESSOR ASKS
What is the input to your classifier?
EXCELLENT ANSWER

The classifier receives a seventeen-dimensional feature vector extracted from the transient response waveform.

PROFESSOR ASKS
What is the output?
EXCELLENT ANSWER

The output is one of the predefined fault classes representing the predicted circuit condition. Why Not Use Images? Good question. Some ML projects use images. We don't. Because our data is electrical signals. **Signals** — Features. Classifier What Makes Our Project Different? Many ML projects look like

Download Dataset

Train Model

Done Our project is much larger.

Circuit Design

Simulation

Dataset Generation

Signal Processing

Feature Engineering

Machine Learning

Evaluation

Engineering Improvements

Dashboard This is why I kept telling you your project is not just an SVM project. It is an engineering project that uses machine learning. Biggest Strength of Our Project Question. What is the biggest strength? Many students answer "SVM." Wrong. Others answer "ANN." Wrong. The biggest strength is the complete pipeline. Because even if tomorrow you replace SVM with

Random Forest

or ANN or XGBoost the rest of the project still works. That is called a modular pipeline. Very important.

Biggest Weakness

Every project has limitations. Our biggest ones are Simulation dataset Limited circuits Limited fault types Feature engineering is manual Needs hardware validation We'll discuss these later.

One Thing You Should Always Remember

Never introduce the project like this We used SVM. Wrong. Always introduce it like this We developed an automated analog circuit fault diagnosis pipeline using machine learning. Pipeline. Not algorithm.

30-Second Project Explanation

If HR asks "Tell me about your project." Answer Our project focuses on automatic fault diagnosis in analog circuits. We first generate circuit data using LTspice, extract meaningful features from transient waveforms, and then train a machine learning model to identify different fault conditions. We further improve the system using feature engineering, error analysis, and engineering constraints, making the diagnosis more reliable. About 30 seconds. Perfect.

2-Minute Project Explanation

If a professor asks "Explain your project." Answer Our objective was to automate analog circuit fault diagnosis using machine learning. We began by designing analog circuits in LTspice and generating transient response waveforms for both healthy and faulty conditions. Since machine learning cannot directly interpret waveforms, we extracted seventeen meaningful features describing the electrical behavior of each signal. These standardized feature vectors were then used to train an optimized Support Vector Machine classifier. Finally, we evaluated the model using accuracy, precision, recall, F1-score, and confusion matrix analysis. Based on the observed errors, we improved the pipeline by introducing additional features such as DC Level and topology-aware post-processing. We later extended the work by comparing the SVM with an ANN using the same preprocessing pipeline.

COMMON MISTAKES

MISTAKE: [X] "Our project is SVM." Wrong. SVM is only one module.

MISTAKE: [X] "Our project is Python." Wrong. Python is only the implementation language.

MISTAKE: [X] "Our project is ANN." Wrong. ANN is only another classifier. The project is the entire pipeline.

REVISION NOTES

Remember these five points.

Problem — Automatic fault diagnosis.

Input — 17 Features.

Output — Fault Class.

Core Idea — Machine Learning predicts faults from waveform features.

Biggest Strength — Complete engineering pipeline.

KNOWLEDGE CHECK

Level 1

- What is the main objective of the project?
- What is the input to the classifier?
- What is the output?
- Why do we need machine learning?
- Why analog circuits?

Level 2

- Why is this not just an SVM project?
- Why is feature extraction necessary?
- What are the seven major modules?
- What makes this project different from a typical ML project?
- Why is the pipeline modular?

Professor Level

- Why should anyone trust a simulation-based dataset?
- Why did you choose machine learning instead of rule-based diagnosis?
- What is the biggest contribution of your work?
- If I remove the classifier, which parts of the project still remain useful?
- What is the single biggest limitation of your project?
- End of Chapter 1

Congratulations. You have completed the first chapter of the Project Bible. This chapter serves as the mental map for everything that follows. Every later chapter—LTspice, interpolation, feature extraction, SVM, ANN, dashboards, and presentation—will simply expand one box of the pipeline we introduced here. Next Chapter: Problem Statement & Motivation (when you type continue).

CHAPTER 2 — Problem Statement & Motivation

Chapter 2 — Problem Statement & Motivation Before Starting

In Chapter 1

we answered What is the project? In this chapter we answer Why was this project built? This is the chapter that makes the professor think "Yes, this problem is worth solving." Many students make the mistake of jumping directly into algorithms. Don't. A good engineer first explains the problem.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] Why this project exists [OK] What problem we are solving [OK] Why analog circuits are difficult [OK] Why manual diagnosis is difficult [OK] Why automation is needed [OK] Why machine learning is suitable [OK] Explain the motivation confidently

Difficulty

Basics 5/5

Theory 5/5

Coding 1/5

Viva Importance 5/5

Let's Begin

Suppose you are working in a company. One day an electronic device stops working. Question. What do you do first? Most people answer Replace the board. Wrong. Engineers first ask Which component failed? That process is called Fault Diagnosis. What is Fault Diagnosis? Very simple. Fault Diagnosis means finding which component inside a circuit has become faulty. Example

Normal Circuit

Works properly.

Faulty Circuit

Not working. Question. Which component failed? That is fault diagnosis.

PROFESSOR ASKS

What is Fault Diagnosis?

EXCELLENT ANSWER

Fault diagnosis is the process of identifying and locating faults within an electronic circuit so that appropriate corrective action can be taken. Simple. Professional. What is a Fault? Question. What exactly is a fault? A fault is any abnormal condition that causes the circuit to behave differently from its intended operation. Example Normal resistor

100 Ω

Fault

Open Circuit

or

150 Ω

because of aging. Both are faults. Types of Faults In electronics faults are generally three types.

1. Open Circuit

Component connection breaks. Example

$$R = \infty$$

No current flows.

2. Short Circuit

Resistance becomes almost zero. Current increases abnormally.

3. Parameter Drift

Component still works but its value changes. Example

100 Ω

130 Ω

Still connected. Still working. But incorrect. Our project mainly studies these types of faults. Why Are Analog Circuits Difficult? Question. Why not just use digital circuits? Let's compare.

Digital Circuit

Output 0 or 1 Easy.

Analog Circuit

Output 0.23 1.74 3.51 2.86. Continuous. Much harder.

PROFESSOR ASKS

Why are analog circuits more difficult than digital circuits?

EXCELLENT ANSWER

Analog circuits produce continuous-valued signals, making faults more difficult to distinguish because even small component variations can significantly alter circuit behavior. Excellent.

Manual Fault Diagnosis

Suppose you have this waveform. Voltage

|
| / \
| / \
|_____/_____ Time

Question. Can you tell whether the resistor or capacitor is faulty? Not easy. Requires experience. Now imagine 1000 waveforms. Can one engineer inspect all of them? Very difficult. Need Automation.

Real Life Example

Imagine an automobile. Modern cars contain hundreds of analog circuits. If one circuit fails, finding the faulty component manually takes time. Automatic diagnosis can identify the probable fault within seconds. Another example Hospital Equipment. If an ECG machine starts giving incorrect readings, manual diagnosis may delay treatment. Automatic diagnosis helps engineers identify the problem quickly. Question Can we simply write rules? Example

If Peak Voltage > 5

Fault A

Question. Will this work? Maybe. Now add 17 features. 16 fault classes. Thousands of combinations. Writing rules becomes almost impossible. Need Machine Learning. Why Machine Learning?

Machine Learning

doesn't require us to manually write hundreds of rules. Instead it learns patterns from data. Think of it like teaching a student. Instead of saying Memorize this rule. We show many examples. Eventually the student learns the pattern.

Machine Learning

works the same way.

PROFESSOR ASKS

Why Machine Learning instead of Rule-Based Systems?

EXCELLENT ANSWER

Rule-based systems become difficult to maintain as the number of fault classes and waveform characteristics increases. Machine learning automatically learns complex relationships from data, making it more suitable for multiclass fault diagnosis.

Important Point

Machine Learning

does not replace electronics. It works because electronics provides meaningful data. Always remember Electronics

Signal Processing

Machine Learning

Not

Machine Learning

alone. Where Does Our Data Come From? Good question.

Machine Learning

needs training data. Question. Where do we get it? Real hardware? Possible. But expensive. Slow. Time-consuming. Instead we use LTspice. We'll study LTspice later. For now remember **Circuit** — LTspice. **Waveform** — Dataset. Another Question

Can Machine Learning

understand waveforms directly? No. **Waveform** — Thousands of points.

Machine Learning

needs numbers. So we convert waveforms into Features. Example Peak 5.2 RMS 3.4

Rise Time

1.8.

17 Features

Now

Machine Learning

can learn.

Motivation Behind Our Project

Question. Why exactly did we build this project? Answer Because we wanted to develop an automatic system that detects faults quickly reduces human effort improves consistency assists engineers serves as a foundation for intelligent diagnostic systems Notice we are helping engineers. Not replacing them. What Makes This Project Useful? Let's compare. Traditional Method **Circuit** — Engineer.

Manual Analysis

Fault

Our Method

Circuit — LTspice

Waveform — Features

Machine Learning

Fault Much faster.

PROFESSOR ASKS

What problem does your project solve?

EXCELLENT ANSWER

The project automates analog circuit fault diagnosis by using machine learning to identify fault conditions from transient response waveforms, reducing the need for manual waveform interpretation.

PROFESSOR ASKS

Why is manual diagnosis difficult?

EXCELLENT ANSWER

Manual diagnosis requires expert interpretation of waveform characteristics and becomes increasingly difficult as circuit complexity and the number of possible fault conditions increase.

One Important Clarification

Our project does NOT repair circuits. It only identifies the fault. Repair is done by the engineer. Don't confuse diagnosis with repair.

Common Misconception

Some students say "Our project detects faults." Not completely correct. Our project classifies faults. The fault already exists. We identify which one it is. That's classification.

One Line That Impresses Professors

Instead of saying We built an SVM model. Say We developed an automated fault classification framework for analog circuits using transient response analysis and machine learning. Sounds much more like engineering.

Real Contribution

Question. What is the actual contribution of this work? Not SVM. Not ANN. The contribution is building a complete automated fault diagnosis framework. Very important.

How This Project Grew

Originally our project looked like **Waveform** — Features. **SVM** — Prediction. Later through our discussions and improvements it became **Waveform** — Interpolation.

17 Features

StandardScaler — Optimized SVM. Dashboard

Error Analysis

DC Level

Topology Rule

ANN Comparison — SHAP.

Final System

Notice how much the project evolved.

COMMON MISTAKES

MISTAKE: [X] Machine Learning finds faults. Wrong. Machine Learning classifies faults based on features.

MISTAKE: [X] Analog circuits are difficult because they are old. Wrong. They are difficult because their signals are continuous.

MISTAKE: [X] The project repairs circuits. Wrong. It identifies the fault.

REVISION NOTES

Remember these five points.

Problem — Manual fault diagnosis is difficult.

Need — Automation.

Solution — Machine Learning.

Input — Features.

Output — Fault Class.

KNOWLEDGE CHECK

Level 1

- What is fault diagnosis?
- What is a fault?
- Why are analog circuits difficult?
- Why do we need automation?
- Why machine learning?

Level 2

- Why not rule-based systems?
- Why not directly use waveforms?
- What is the contribution of the project?
- What is the difference between diagnosis and repair?
- Why is simulation useful?

Professor Level

- Why did you choose analog circuits instead of digital circuits?
- If machine learning is so powerful, why do you still need feature engineering?
- Why is fault classification a supervised learning problem?
- Could your framework work on real hardware? What changes would be needed?
- What assumptions does your project make about the availability of labeled fault data?
- End of Chapter 2

You now understand why the project exists, what engineering problem it solves, and why machine learning is an appropriate solution. From here onward, we stop discussing why and begin discussing how. The next chapter is where we define the specific objectives of the project and show how they naturally lead into the complete implementation pipeline.

CHAPTER 3 — Project Objectives

Chapter 3 — Project Objectives Before Starting

In Chapter 1

we answered What is the project?

In Chapter 2

we answered Why was this project built? Now we answer What exactly were we trying to achieve? This chapter is very important because many professors ask "What were your objectives?" Most students simply read bullet points. Don't. After this chapter you'll understand why every objective exists.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] What were the objectives? [OK] Why each objective is necessary? [OK] Why are they arranged in this order? [OK] Which objective is the most important? [OK] Explain objectives in viva.

Difficulty

Basics 5/5

Theory 5/5

Coding 1/5

Viva Importance 5/5

Let's Begin

Suppose I ask Why did we make this project? You already know. Automatic Fault Diagnosis. Good. Now Question. How do we actually reach that goal? Can we directly train SVM? No. There are many steps before SVM. Those steps become our objectives. Think Like Building a House Suppose you want to build a house. Question. Can you start with painting? No. Need **Land** — Foundation. **Walls** — Roof. Painting Similarly our project also follows a sequence. Complete Objective Flow Design Circuit

Generate Dataset

Extract Features

Train Machine Learning Model

Evaluate Model

Improve Model

Congratulations. These are the six major objectives of our project. Now let's study each one.

Objective 1 — Design Analog Circuits

Question. Can we build a machine learning model without data? No. Question. Can we get data without circuits? Again No. Therefore the first objective is Design the circuits.

PROFESSOR ASKS

Why design circuits first?

EXCELLENT ANSWER

Machine learning requires data, and the data originates from analog circuit simulations. Therefore, circuit design is the first step of the complete pipeline.

Objective 2 — Generate Dataset

Now we have circuits. Question. Can SVM train using circuits? No. Need data. So we simulate the circuits. **Circuit** — LTspice. **Waveform** — Dataset. Simple.

PROFESSOR ASKS

Why generate a dataset?

EXCELLENT ANSWER

Because supervised machine learning requires labeled examples for training and testing. Notice The keyword is

Labeled Examples

Remember this.

Objective 3 — Feature Extraction

Now Question. Can SVM understand waveforms? Imagine this waveform. Voltage



Question. Can SVM read this? No. Need numbers. Example

Peak = 5.4

RMS = 3.8

Rise Time = 2.1

. Now SVM understands.

PROFESSOR ASKS

Why Feature Extraction?

EXCELLENT ANSWER

Feature extraction converts complex waveforms into meaningful numerical representations suitable for machine learning. Excellent.

Small Analogy

Think of a teacher. Suppose teacher wants student marks. Question. Will teacher keep answer sheets or final marks? Marks. Same idea. **Waveform** — Feature. Like

Answer Sheet

Marks Much easier.

Objective 4 — Train Machine Learning Model

Now Question. We have 17 Features. Can they predict faults themselves? No. Need a classifier. Initially our classifier was SVM. Later we added ANN. Simple.

PROFESSOR ASKS

Why train a classifier?

EXCELLENT ANSWER

The classifier learns the relationship between the extracted features and the corresponding fault classes so that it can classify unseen samples. Notice One important word

Unseen Samples

Machine Learning

must work on data it has never seen before.

Objective 5 — Evaluate Performance

Question. Suppose model gives 80% accuracy. Question. Is it good? Maybe. Question. How do we know? Need Evaluation. We calculate Accuracy Precision Recall F1 Score Confusion Matrix Now we know how good our model really is.

PROFESSOR ASKS
Why not only accuracy?
EXCELLENT ANSWER

Accuracy provides only an overall measure of performance. Precision, recall, F1-score, and confusion matrix provide class-wise evaluation and identify weaknesses in the classifier.

Objective 6 — Improve the System

This is where our project became different. Most students stop here. **Train** — Accuracy. Done We didn't. We continued. Question. Suppose

Confusion Matrix

shows RC_Open being confused. Question. Should we simply accept it? No. We investigate. We ask Why? Then we improve. Example

Confusion Matrix

RC_Open confused — Study waveform

Add DC Level — Retrain

Improved Accuracy

This is called Engineering.

PROFESSOR ASKS
What improvement did you make?
EXCELLENT ANSWER

Error analysis revealed that the existing features did not sufficiently capture steady-state behavior for certain fault classes. Therefore, the DC Level feature was introduced, improving classification performance.

Another Improvement

Question. Suppose classifier predicts an impossible topology. Should we accept it? No. Need Topology Rule. Again Improvement. Notice our project was never just training SVM. It was **Observe** — Analyze. **Improve** — Validate. Very important. Which Objective is the Most Important? Many students answer Training. Wrong. Others answer SVM. Wrong. The most important objective is Building the complete diagnostic pipeline. Because if tomorrow you replace SVM with ANN the project still works.

PROFESSOR ASKS
Which objective is the most important?
EXCELLENT ANSWER

The most important objective is developing a complete automated fault diagnosis pipeline because the individual machine learning model can later be replaced or improved without changing the overall framework. Excellent. Why Are Objectives in This Order? Question. Can we evaluate before training? No. Can we train before extracting features? No. Can we extract features before generating data? No. Everything depends on the previous step. Exactly like building a house. Complete Objective Pipeline Circuit Design

Dataset Generation

Feature Extraction

Machine Learning

Evaluation — Improvement.

Final System

Remember this. It is the backbone of the project.

One Important Thing

Notice something. There is NO objective called Use SVM. Why? Because SVM is only one possible solution. Tomorrow replace it with Random Forest. Objectives remain exactly the same. This is why objectives should always be general.

COMMON MISTAKES

MISTAKE: [X] **Objective** — Use Python. Wrong. Python is a tool. Not an objective.

MISTAKE: [X] **Objective** — Train SVM. Wrong. SVM is one implementation. Not the project objective.

MISTAKE: [X] **Objective** — Get 90% Accuracy. Wrong. Accuracy is a result. Not an objective.
Presentation Tip During presentation don't read the objectives. Instead say To solve the problem systematically, we divided the work into six major objectives. Then explain each one in one sentence. Very professional.

REVISION NOTES

Remember these six objectives.

Design Circuits — Need source of data.

Generate Dataset — Need training samples.

Extract Features — Convert waveforms into numbers.

Train Model — Learn patterns.

Evaluate — Measure performance.

Improve — Make the system better.

KNOWLEDGE CHECK

Level 1

- Why is circuit design the first objective?
- Why do we need a dataset?
- Why feature extraction?
- Why evaluation?
- Why improvement?

Level 2

- Why are objectives arranged in this order?
- Why isn't "Use SVM" an objective?
- Why isn't "Get high accuracy" an objective?
- What makes our objectives different from a basic ML project?
- Which objective contributes most to the engineering value of the project?

Professor Level

- Could this project still work if SVM were replaced with another classifier? Explain.
- Why is feature extraction an objective rather than an implementation detail?
- Which objective required the most engineering decisions in your project?
- If you had real hardware data instead of simulations, which objectives would change and which would remain the same?
- What objective distinguishes your project from a standard machine learning assignment?
- End of Chapter 3

Congratulations. Now you understand not just what the objectives are, but why each one exists and how they connect into a complete engineering workflow.

Author's Note (Something I want to improve) After writing these first three chapters, I realized something. I think I can make this Bible even better than our chat. Starting from Chapter 4, I'm going to add one more section at the end of every chapter: "Memory Tricks" For example: Project Pipeline CD → DG → FE → ML → EV → IM Circuit Design Dataset Generation Feature Extraction Machine Learning Evaluation Improvement Or small mnemonics and quick recall techniques that will help you remember concepts before your viva or interview. I think by the time we finish all ~70 chapters, this won't just be a project guide—it will be a complete mastery guide for your project.

CHAPTER 4 — Complete Project Methodology

Chapter 4 — Complete Project Methodology (The Heart of the Project) Before Starting Till now we have learned

Chapter 1

What is the project?

Chapter 2

Why was it built?

Chapter 3

What are our objectives? Now we study the most important chapter of the entire project. This chapter is the backbone. If you understand this chapter, you understand almost 60% of the project.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] Explain the complete project from start to finish. [OK] Explain every stage. [OK] Explain why every stage is necessary. [OK] Draw the project flow on a whiteboard. [OK] Answer methodology questions in viva.

Difficulty

Basics 5/5

Theory 5/5

Coding 2/5

Viva Importance 5/5

Let's Begin

Suppose your professor asks Explain your complete project. Most students panic. They jump directly to SVM. Wrong. Never start with SVM. Instead always start with the complete pipeline.

Our Entire Project

Everything can be summarized using this single diagram.

Circuit Design

LTspice Simulation

Transient Waveforms

Interpolation

Feature Extraction

17 Features

StandardScaler

Machine Learning

Prediction — Evaluation.

Error Analysis

Improvement

Final System

Congratulations. This is the entire project. Everything else is just explaining each box. Let's Understand Every Box Step 1 Circuit Design Question. Can we generate waveforms without circuits? No. Everything begins here. We first design analog circuits inside LTspice. Example RC Circuit RL Circuit RLC Circuit Question. Why? Because we need something to simulate.

PROFESSOR ASKS

Why do we design circuits first?

EXCELLENT ANSWER

The circuit is the source of all data. Without designing the circuit, we cannot perform simulations or generate training data.

Step 2 — LTspice Simulation

Now we have circuits. Question. Can machine learning read circuits? No. Need waveforms. So LTspice simulates the electrical behavior. Example Voltage

|

| ^ | / \

|___/___ Time

Now we have signals.

PROFESSOR ASKS

Why LTspice?

EXCELLENT ANSWER

LTspice accurately simulates analog circuit behavior and allows us to generate labeled data efficiently without requiring physical hardware.

Step 3 — Dataset Generation

One waveform is not enough.

Machine Learning

needs many examples. Example Healthy Healthy Healthy RC Open RC Short RL Open RLC Fault. Each waveform gets a label. Question. Why? Because we are doing Supervised Learning.

PROFESSOR ASKS

What is supervised learning?

EXCELLENT ANSWER

Supervised learning is a machine learning approach in which the model learns from labeled training data consisting of input-output pairs. Excellent.

Step 4 — Raw Waveforms

Now Question. Can we directly give waveforms to SVM? No. Imagine

Waveform A

850 points

Waveform B

1027 points Question. Can SVM compare them? No. Need Preprocessing.

Step 5 — Interpolation

Interpolation makes all waveforms consistent. Example **850** — 1000. **1027** — 1000. Now every waveform has 1000 points. Question. Why? So feature extraction becomes fair.

PROFESSOR ASKS

Does interpolation create new information?

EXCELLENT ANSWER

No. Interpolation only estimates intermediate values to produce a uniform representation. Very important.

Step 6 — Feature Extraction

Question. Can SVM understand 1000 voltage points? Technically yes. But not efficiently for our project. Instead we summarize the waveform. Example Peak RMS Mean

Rise Time

Entropy.

17 Features

Question. Why? Because

Machine Learning

works better when important information is represented as meaningful numerical features.

PROFESSOR ASKS

What is feature extraction?

EXCELLENT ANSWER

Feature extraction converts raw waveform data into compact numerical descriptors that preserve important characteristics for machine learning.

Step 7 — Feature Vector

Now each waveform becomes something like [5.4, 3.2, 0.84, 1.73,. 4.9] Exactly 17 numbers. Question. What do we call this? Feature Vector. Remember this word. Very important.

PROFESSOR ASKS

What is a Feature Vector?

EXCELLENT ANSWER

A feature vector is the numerical representation of one waveform obtained after feature extraction.

Step 8 — StandardScaler

Question. Suppose

Peak Voltage

is 520

Rise Time

is 0.002 Question. Can SVM treat them fairly? No. Peak dominates. Need Scaling. StandardScaler makes every feature have similar scale. Now SVM learns properly.

PROFESSOR ASKS

Why StandardScaler?

EXCELLENT ANSWER

StandardScaler standardizes feature values so that large numerical features do not dominate smaller ones during model training.

Step 9 — Machine Learning

Finally we reach SVM. Notice something. SVM is the ninth step. Not the first. Very important. **Input** — 17 Features. **Output** — Fault Class. Initially we use SVM. Later ANN.

PROFESSOR ASKS

Why SVM?

EXCELLENT ANSWER

SVM performs well on structured numerical datasets with engineered features and provides good generalization for relatively small datasets.

Step 10 — Prediction

Question. What does SVM predict? Voltage? No. Features? No. It predicts the Fault Class. Example RC_Open or RL_Normal Simple.

Step 11 — Evaluation

Question. Suppose accuracy is 82%. Question. Finished? No. Need more analysis. We calculate Accuracy Precision Recall F1

Confusion Matrix

Now we know how well our model performs.

PROFESSOR ASKS

Why multiple metrics?

EXCELLENT ANSWER

Different metrics evaluate different aspects of classifier performance. Together they provide a more complete understanding than accuracy alone.

Step 12 — Error Analysis

This is where our project became interesting. Question. Suppose

Confusion Matrix

shows **RC_Open** — Misclassified. Question. Should we ignore it? No. We investigate. Why? This is called Error Analysis.

Step 13 — Improvement

Error Analysis

shows steady-state information is missing. Question. Solution? Add DC Level. Retrain. Performance improves. Again Engineering. Not Guesswork.

Step 14 — Topology Rule

Question. Suppose classifier predicts an impossible circuit topology. Should we trust it? No. We already know the topology. So

Topology Rule

corrects physically impossible predictions.

PROFESSOR ASKS

Is this cheating?

EXCELLENT ANSWER

No. The topology rule incorporates engineering knowledge after prediction. It does not change the trained classifier.

Step 15 — Final System

After all improvements our pipeline becomes **Circuit** — Waveform. **Features** — StandardScaler. **SVM** — Evaluation.

Error Analysis

DC Level

Topology Rule

Final Prediction

This is our final system.

One Very Important Observation

Notice the project never goes directly from **Waveform** — SVM. Instead there are many engineering stages. That is why this project is much stronger than a normal ML assignment. Professor's Favourite Question Which stage is the most important? Most students say SVM. Wrong. The correct answer is Feature Engineering. Because if your features are poor, even the best classifier cannot perform well. Remember this.

Another Professor Question

Can we replace SVM? Yes. Replace SVM with ANN. Or Random Forest. Or XGBoost. Everything else remains the same. This proves our pipeline is modular. Why Is This Called a Pipeline? Question. Why not simply call it a project? Because every stage feeds the next stage. Exactly like a water pipeline.

Stage A

Stage B

Stage C

Stage D

Output of one stage becomes input to the next. That is a pipeline.

COMMON MISTAKES

MISTAKE: [X] **Project** — SVM. Wrong.

MISTAKE: [X] **Machine Learning** — first step. Wrong.

MISTAKE: [X] **Evaluation** — only Accuracy. Wrong.

MISTAKE: [X] Stop after training. Wrong. Need analysis and improvement.

REVISION NOTES

Remember this flow. **Circuit** — Simulation. **Waveforms** — Interpolation. **Feature Extraction** — 17 Features. **StandardScaler** — SVM. **Prediction** — Evaluation. Improvement If you remember this one diagram, you can explain the entire project.

MEMORY TRICK

Remember this sequence: $C \rightarrow S \rightarrow W \rightarrow I \rightarrow F \rightarrow S \rightarrow M \rightarrow P \rightarrow E \rightarrow I$ Expand it: C = Circuit Design S = Simulation W = Waveforms I = Interpolation F = Feature Extraction S = StandardScaler M = Machine Learning P = Prediction E = Evaluation I = Improvement Before every viva, repeat this sequence three times. You'll never forget the methodology.

KNOWLEDGE CHECK

Level 1

- What is the first step of the project?
- Why do we use LTspice?
- Why do we perform interpolation?
- Why do we extract features?
- Why do we use StandardScaler?

Level 2

- Why is SVM not the first stage?
- What is a feature vector?
- Why is the pipeline modular?
- Why do we perform error analysis?
- Why was the DC Level feature added?

Professor Level

- If I remove interpolation, what problems could arise later in the pipeline?
- Which stage contributes most to the classifier's success, and why?
- How does engineering knowledge influence the pipeline beyond machine learning?
- If you replaced SVM with ANN, which stages would remain unchanged?
- Explain the complete methodology on a whiteboard without referring to your slides.
- End of Chapter 4

Congratulations. You now understand the complete project methodology. This chapter is the center of the entire Project Bible. Every chapter that follows—LTspice, dataset generation, interpolation, feature extraction, SVM, evaluation, ANN—will simply zoom into one box of this pipeline and explain it in depth. Next Chapter: Analog Electronics Fundamentals (RC, RL, RLC circuits and the electronics behind your project). This is where we'll connect the machine learning with the actual electronics, which is often where professors start asking deeper technical questions.